

Lab 3: Feature Attribution and the Responsibility Gap

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Lab Report: Who Bears Responsibility?

Google's What-If Tool (WIT) is built to expose a global view of which features drive predictions overall (the Features tab) and a local view of why a single prediction was made (Wexler et al., 2020). This income-classification demo does not expose the SHAP-style attribution bars; the assignment describes only Nearest Counterfactual, partial dependence, and manual editing. That gap is itself informative: explainability is not one fixed output every model provides, but whatever a tool's backend was actually built to surface, and a practitioner has to know which kind of explanation is in front of them before trusting it.

Step 2 asks which feature carries the highest variance. In this session's Features tab (Figure 1), capital-gain is the clear outlier: $SD = 9,298.31$ against a mean of \$1,470, roughly $27\times$ the next-highest feature, capital-loss ($SD = 354.58$). Ninety percent of data points report exactly zero; the remainder range up to \$100,000 (Lundberg & Lee, 2017, report a similar skew in independent SHAP analyses of this dataset). That skew creates three risks: brittleness, since a model tuned to a rare nonzero subgroup can swing sharply on inputs almost nobody presents; bias, since access to capital gains is not evenly distributed across the population being scored; and poor generalization, since 90% of the training data carries no signal on this feature at all, leaving the model to fit noise in a thin 10% subgroup.

Step 3 asks for a case where high education is still paired with a predicted income of $\leq \$50K$. Datapoint 259 (Figure 2) fits that directly; a 72 year old with Prof-school training (education-num = 15, one point short of the dataset's ceiling of 16), self-employed in a Prof-specialty occupation, working 20 hours/week, never married. Despite that education level, both models lean toward $\leq \$50K$ ($P(>50K) = 0.344$ and 0.372). Toggling Nearest Counterfactual (Figure 3) surfaces datapoint 382, the closest real case classified the other way, and the prediction climbs to a confident $> \$50K$ (0.943 and 0.778) but nine values move at once to get there: age ($72 \rightarrow 58$), capital-loss ($1,429 \rightarrow 1,485$), education (Prof-school $\rightarrow$ Doctorate), hours-per-week ($20 \rightarrow 35$), marital-status (Never-married $\rightarrow$ Married-AF-spouse), occupation (Prof-specialty $\rightarrow$ Exec-managerial), relationship (Not-in-family $\rightarrow$ Wife), and sex (Male $\rightarrow$ Female).

With nine simultaneous changes, Figure 3's counterfactual cannot show which feature outweighed education, so five single feature edits were run on datapoint 259, changing one value at a time with education and everything else untouched (Table 1); the Nearest Counterfactual toggle remained on throughout, so Figures 4 and 5 also display WIT's automatically paired

comparison datapoint (382, 499) in the Infer panel, though only datapoint 259's own column is reported here. Sex alone moved both models only modestly, and they disagree on which direction it even moves the score; neither prediction flips. Occupation and hours-per-week alone were similarly modest and non-flipping. Marital-status alone crossed the 0.5 threshold on both models from changing that single field. Capital-gain alone (Figure 5) was the largest effect of all larger than marital-status, and larger than the full nine-feature counterfactual in Figure 3 pushing both models to near-certainty on >\$50K from a single \$7,000 edit. That result directly confirms Step 2's global finding: capital-gain's extreme, mostly-zero skew means the model treats any nonzero value as an almost decisive signal.

Table 1

Effect of Single-Feature Edits on Datapoint 259's Predicted Probability of Earning >\$50K

Feature Changed	Value Change	Model 1 P(>50K)	Model 2 P(>50K)	Flips Prediction?
Baseline	— (none)	0.344	0.372	—
Sex	Male → Female	0.430	0.326	No
Occupation	Prof-specialty → Exec-managerial	0.378	0.415	No
Hours-per-week	20 → 35	0.426	0.406	No
Marital-status	Never-married → Married-civ-spouse	0.603	0.710	Yes (both)
Capital-gain	\$0 → \$7,000	0.983	0.798	Yes (both, largest)

That contrast sharpens rather than weakens the proxy-discrimination concern raised earlier. Capital-gain's dominance is a wealth-access story, not a protected-class one; it is the largest lever here, but it is not evidence of gender proxying. Marital-status is the one that is: a naive audit testing only “does this model use sex” flipping sex alone would report a small, inconsistent effect and stop. But marital-status is not independent of sex; this dataset cannot record someone as “Wife” without also recording them as female, and U.S. marital-status categories have long tracked gendered household roles. A model leaning on marital-status can reproduce a gendered pattern without ever using sex directly the failure of fairness through unawareness Barocas and Selbst (2016) describe: testing a protected attribute alone misses a correlated field carrying the same information.

These experiments show the models' statistical behavior, not a causal account of datapoint 259's real income. Flipping marital-status does not mean marrying would raise this person's actual earnings, and flipping sex does not mean gender caused their outcome each shows only what the classifiers output under a different input. A legitimate human explanation would draw on this person's career, health, and choices no manual edit can access. This is evidence about decision boundaries, not the world conflating the two is exactly what this step is meant to expose.

Who, then, is responsible for outcomes like these? Matthias's (2004) “responsibility gap” is the uncomfortable starting point; once an output emerges from learned patterns rather than hand authored rules, no single human fully authored the decision, and blame frameworks strain to attach to anyone. But a gap in traceability is not a gap in accountability. The data provider bears responsibility for what these features encode; capital-gain (Step 2) marks access to investable assets, unevenly distributed by race and generational wealth; marital-status (Step 3) shown to single handedly flip both models is inseparable from sex in this dataset, exactly the proxy problem Barocas and Selbst (2016) describe. The developer bears responsibility for deploying a model built on those features without auditing whether they stood in for wealth and gender rather than labor the due diligence floor Diakopoulos (2016) describes. The system itself cannot be responsible; it has no capacity for judgment or remedy, only pattern matching (System I) dressed as reasoned decision (System II; Meert et al., 2025). Locating blame in “the system” launders avoidable choices into apparent inevitability.

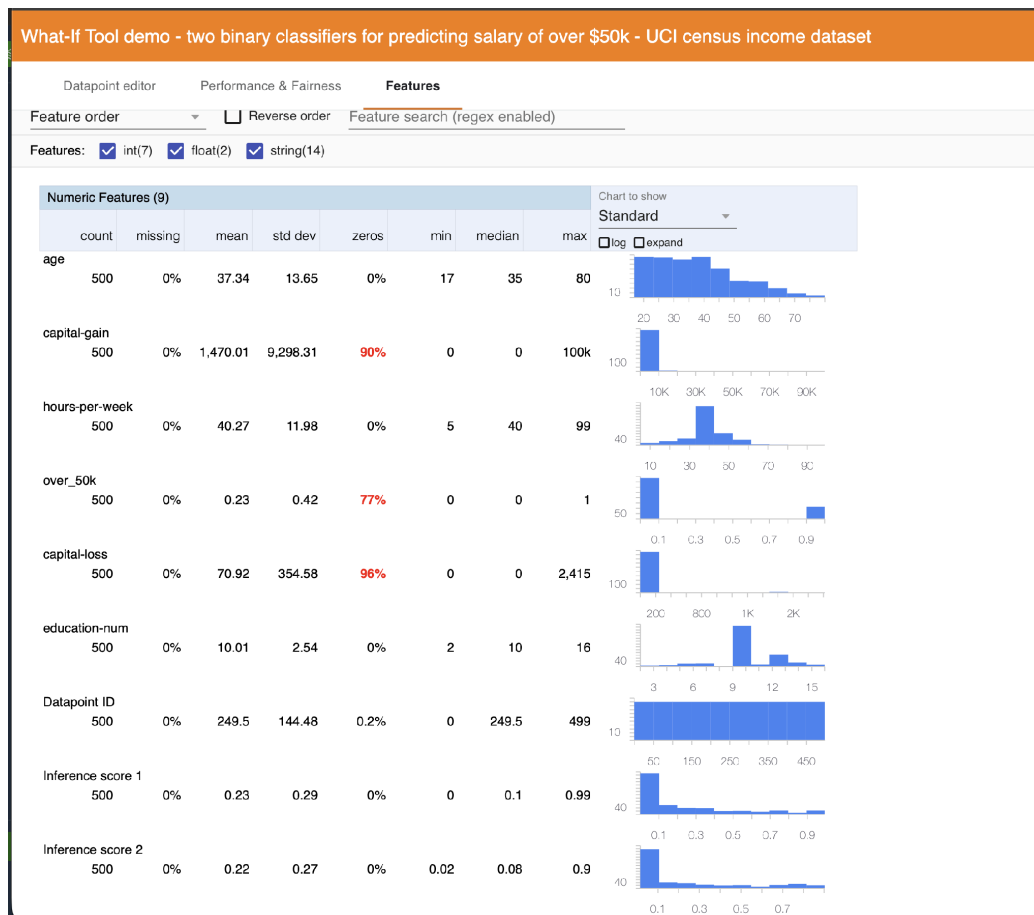


Figure 1. Numeric features table from the WIT Features tab: capital-gain's standard deviation (9,298.31) dwarfs every other feature (Step 2).

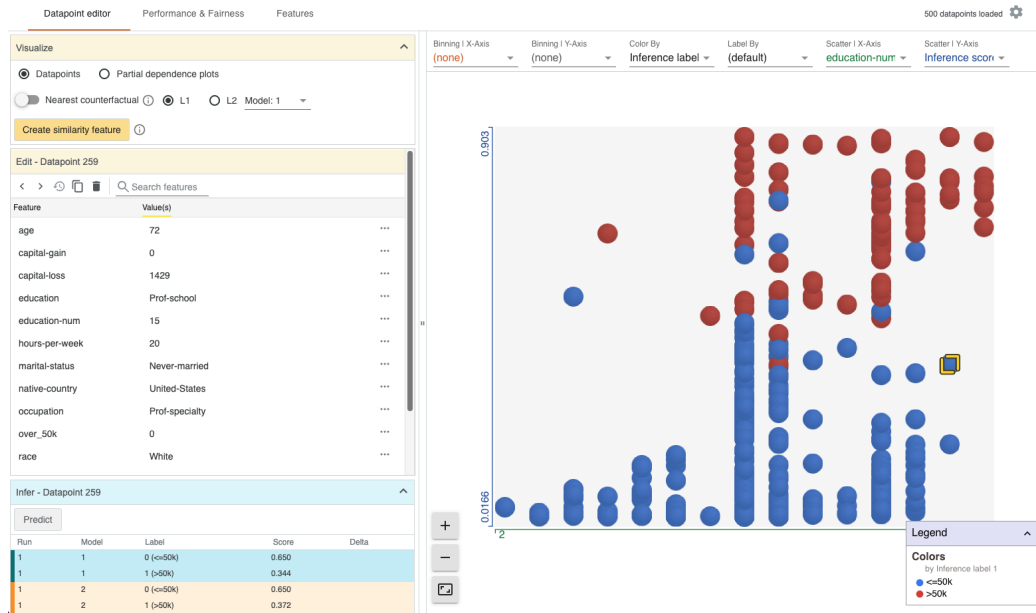


Figure 2. Datapoint 259 before any edit: age 72, Prof-school (education-num = 15), self-employed Prof-specialty, 20 hrs/week, never married — yet both models predict $\leq \$50K$ (Step 3, original datapoint and prediction).

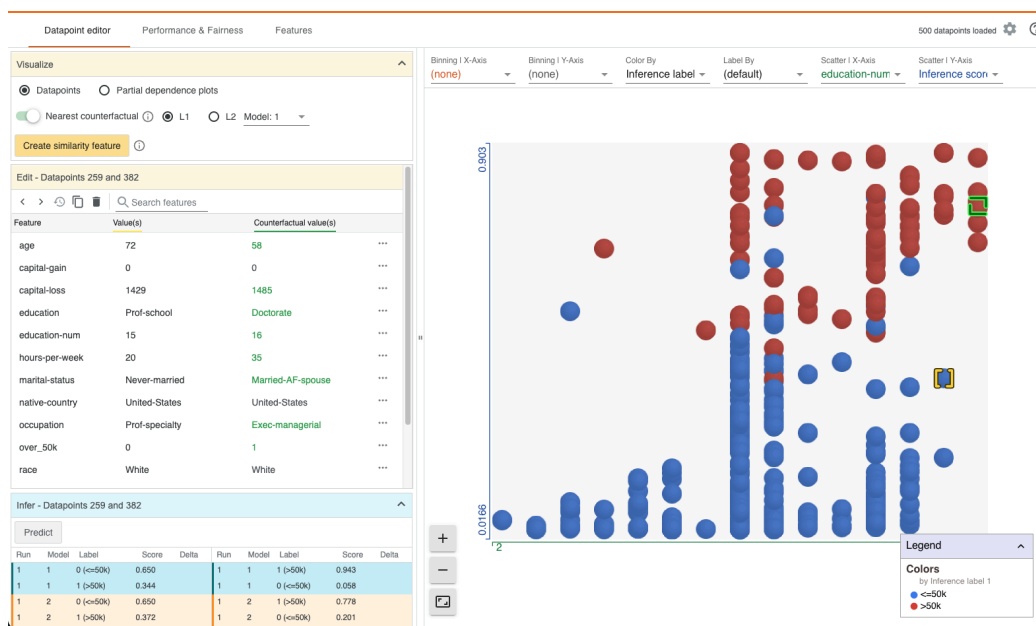


Figure 3. Nearest Counterfactual comparison for datapoint 259: nine features differ from the closest oppositely-classified real datapoint (382), whose predictions climb to 0.943 and 0.778 — note marital-status, relationship, and sex changing together (Step 3, counterfactual and resulting change in prediction).

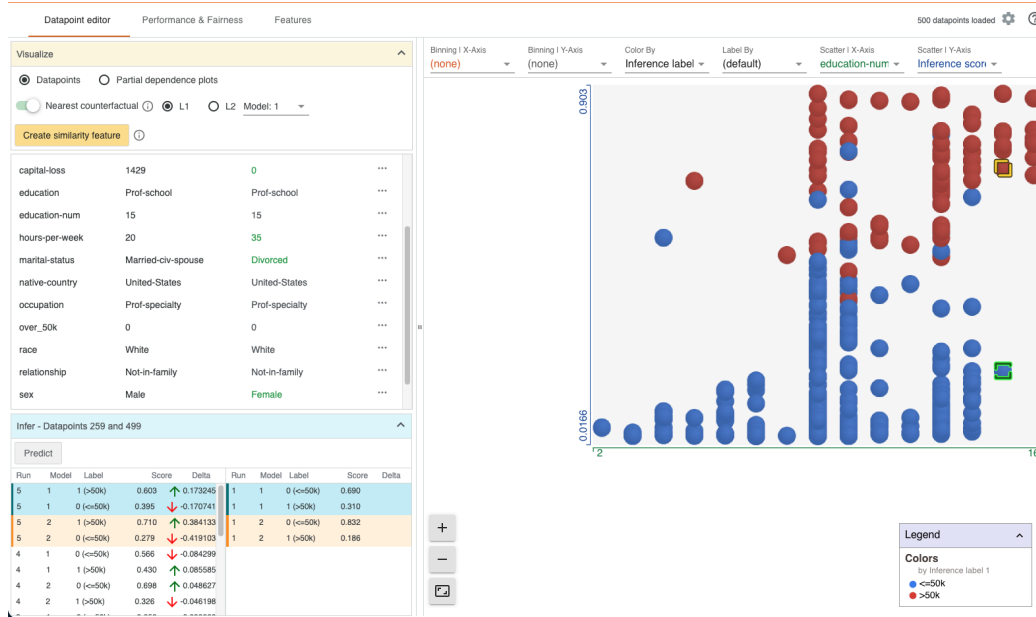


Figure 4. Manual single-feature edit on datapoint 259: *marital-status* alone (Never-married→Married-civ-spouse), all other fields at their original values. Both models flip to predicting >\$50K (0.603 and 0.710) — the largest isolated effect among the four demographic/labor features tested (Step 3, manual feature change and resulting change in prediction).

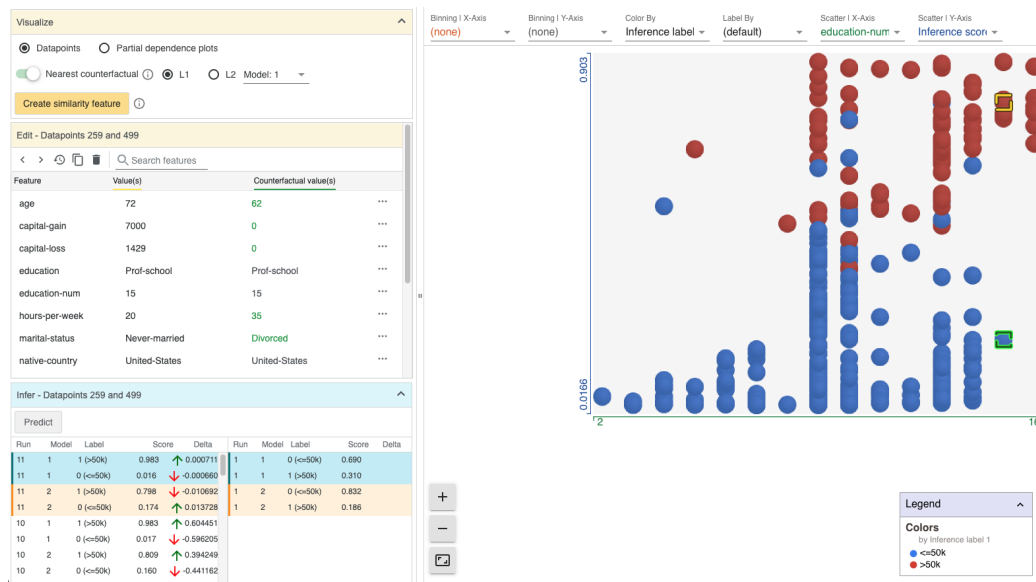


Figure 5. Manual single-feature edit on datapoint 259: *capital-gain* alone (\$0→\$7,000), all other fields at their original values. Both models jump to near-certain >\$50K predictions (0.983 and 0.798) — the largest isolated effect of any feature tested (Step 3, manual feature change and resulting change in prediction).

Final Argument: Explainability Is a Prerequisite for Accountability

Explainability is not a courtesy added to machine learning after the fact it is the precondition that makes accountability possible at all. Diakopoulos (2016) defines accountability as requiring an identifiable party with the authority to address adverse individual or societal effects in a timely fashion. That identification is impossible without first knowing which input produced the harmful output, which is exactly the mapping SHAP values are built to provide; a decomposition of a prediction into the marginal contribution of each feature, and therefore a distribution of responsibility across the inputs a model actually relied on (Lundberg & Lee, 2017). Tools like the What-If Tool operationalize this by putting that decomposition in front of a human who can ask “why,” compare counterfactuals, and audit for exactly the kind of proxy discrimination this lab uncovered (Wexler et al., 2020). Without that layer, a person harmed by a decision has no purchase point: they cannot contest a decision they cannot see inside, and an institution cannot correct a failure mode it cannot name.

However, it should be emphasized that an explanation alone does not guarantee the right outcome. According to Rudin (2019), the act of revealing the logic behind a black box at a late stage fundamentally differs and is inferior compared to the practice of a model transparent to the users being a natural outcome of the design; SHAP tells us that with a training model our predictions were correct while at the same time showing that fairer alternatives were not as accurate. Accountability built entirely on post hoc explanation risks becoming a performance of transparency rather than its substance narrating a decision instead of constraining it. Selbst et al. (2019) push further with their warning about the “abstraction trap”; treating a feature attribution report as a self contained, portable fact about fairness, detached from the social context a model operates in, can manufacture a false sense that the accountability problem has already been solved the moment a SHAP chart is produced.

Taken together, these caveats sharpen the thesis rather than undermine it. Explainability is necessary but not sufficient which is exactly what “prerequisite” means. No one can be held accountable for a decision that cannot be traced, but tracing it is only the first step; someone with actual authority still has to act on what the trace reveals; refuse the deployment, retrain the model, or drop the offending feature. In the census income model examined here, a counterfactual comparison could not decompose any one feature's exact share of the blame the way a true SHAP value would, but single feature manual edits did make capital-gain's and marital-status's outsized roles visible and arguable, one at a time which is the only condition under which a developer, a regulator, or a court could ever be expected to intervene. An unexplainable model forecloses that conversation before it starts. That is the harm explainability guards against, and the reason accountability cannot be built without it.

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